CHAPTER 19

# First and Second Generation Production of Bio-Adipic Acid

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## Abstract

Adipic acid is an important platform chemical with a turnover of 2.5 million tons per year. It is mainly applied as an intermediate reactant for the production of nylon-6,6, a world-leading polymer in the textile and the automobile industry. Due to a relatively high energy demand and greenhouse gas emission, inherently linked to the traditional petrochemical production, great potential exists for more sustainable bio-based alternatives. In pioneering developments, sugars, fats, and alkanes are used as novel feedstock for the production of bio-adipic acid. Recent approaches that recruit small

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aromatic compounds from lignin-based materials seem even more promising. Based on life cycle assessment, such aromatics from lignin reduce energy demand and carbon dioxide equivalent emission significantly, in comparison to the conventional petrochemical process. Lignin-rich streams are available in huge amounts as waste streams of low value that accumulate during processing of lignocellulosic biomass. Recent development of superior producing microorganisms and integrated bioprocesses are enabling technologies toward future industrial production of bio-adipic acid, which is highlighted in this chapter.

## 19.1 INTRODUCTION

Since the development of nylon-6,6, one of the most important industrial polymers worldwide, almost 100 years ago, adipic acid has been an important chemical, as it is the main precursor for this product. Still today, about 85% of the total market of about 2.5 million tons adipic acid per year is used for the production of nylon-6,6 in a copolymerization process with hexamethylenediamine. In addition, a range of other applications for adipic acid have evolved during the past decades involving polyurethane (5%) and adipate esters (4%) in the automotive and the textile industry as two important examples. Additionally, adipic acid is used in the food, pharmaceutical, and agrochemical market. From early on, adipic acid has been produced from petrochemicals. Today, the market leaders for the production of adipic acid are Invista (21%) and Ascend (13%) in the United States, Rodia (18%), as part of the Solvay group in Belgium, and BASF (9%) in Germany. The major production route involves nitric acid oxidation of cyclohexanone and cyclohexanol at 250°C and 55 bars, which are both obtained from naphtha-based benzene (Draths and Frost, 1993). The two conventional feedstock, benzene and nitric acid, are derived from oil and natural gas, respectively. Alternative petrochemical routes are the carbonylation of butadiene and the oxidation of cyclohexane with hydrogen peroxide. In the latter case, four moles hydrogen peroxide are consumed per mole adipic acid. The prices for butadiene and hydrogen peroxide are, however, high (Sato et al., 1998). Furthermore, petrochemical production for adipic acid requires an enormous amount of energy and causes significant greenhouse gas emissions (Vink et al., 2003). The cumulative energy demand (CED) for petrochemical adipic acid is as high as 100 GJ/ton, which includes raw material and process energy input (Patel, 2003). Moreover, the nitric acid oxidation directly yields nitrous oxide ($N_2O$) in addition to adipic acid (Shimizu et al., 2000). This is of major environmental concern, because $N_2O$ represents a greenhouse gas, 310 times more severe as carbon dioxide ($CO_2$).

In order to become less dependent on oil and to reduce the ecological footprint of adipic acid production, the use of biomass as alternative feedstock is promising. However, such bio-based routes to a novel bio-based adipic acid (bio-adipic acid) have to compete with conventional processes with regard to the economic efficiency. In this regard, the low price of adipic acid about €1500/ton (ICIS Pricing) and the fully depreciated petrochemical plants are challenging for new bio-based processes, as this requests for high titers, yields, and productivities. Consequently, low-cost

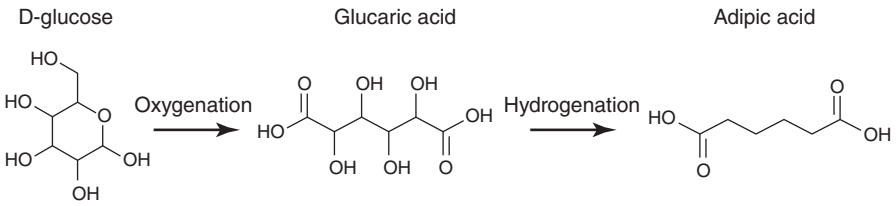



**FIGURE 19.1** Rennovia process for production of bio-adipic acid from glucose via two chemical catalytic steps (Boussie et al., 2010; de Guzman, 2010; Dapsens et al., 2012).

feedstocks have to be used in an efficient and integrated way in order to enable good earnings and to reduce ecological footprints (Hatti-Kaul et al., 2007).

As demonstrated by several early studies, the achievable purity of bio-adipic acid meets the requirements for the fabrication of nylon-6,6 fibers, and resins (Schmidt and Knackmuss, 1984; Mizuno et al., 1988; Draths and Frost, 1993). Thus, future commercial bio-adipic acid might work as a drop-in product. This makes bio-based, chemical or biochemical alternatives particularly attractive: The product is compatible with existing processes downstream of the value chain to the industrial product, so that a large market can be addressed and growth is foreseen. First attempts have already demonstrated production of bio-adipic acid from glucose by a purely chemical process. Shortly, glucose as starting material is converted to bio-adipic acid by two chemical steps using metal catalysts (Boussie et al., 2010; de Guzman, 2010; Dapsens et al., 2012) (Figure 19.1).

First, glucose is oxygenated to the intermediate glucaric acid, which is subsequently hydrogenated to bio-adipic acid. Both conversion steps, however, require high temperature and pressure and the use of supporting solvents. In addition to chemical production, bio-adipic acid can be formed through occurring biosynthetic routes from benzene, found naturally in different microorganisms (Cheng et al., 2002). In particular, this is a good starting point toward *de novo* biosynthetic processes, as described below in more detail. Meanwhile, the use of metabolically engineered microorganisms even allows production of bio-adipic acid from glucose under nature-friendly conditions (Figure 19.2).

In recent years, corresponding biotechnological processes from renewable feedstocks have been developed. Today, both routes—chemical conversion and biochemical conversion of glucose—are about to be commercialized. Beyond glucose, second generation production is on the horizon which involves non-food raw materials such as fats, alkanes, pentose sugars, as well as small aromatics from lignin (Figure 19.3).

Particularly, lignin is an interesting resource, because it is a cheap and widely available starting material (Tuck et al., 2012). Moreover, the chemical structures of the major aromatics are, in contrast to other potential substrates, similar to the desired products, which allows straightforward bioconversion. Huge amounts of lignin accumulate as side-product from lignocellulosic bioethanol. Lignin waste displays up to 25% of the initial resource (de Wild et al., 2010). Until now, it is mainly burned for the generation of electricity. In recent years, thermochemical and lignolytic enzyme

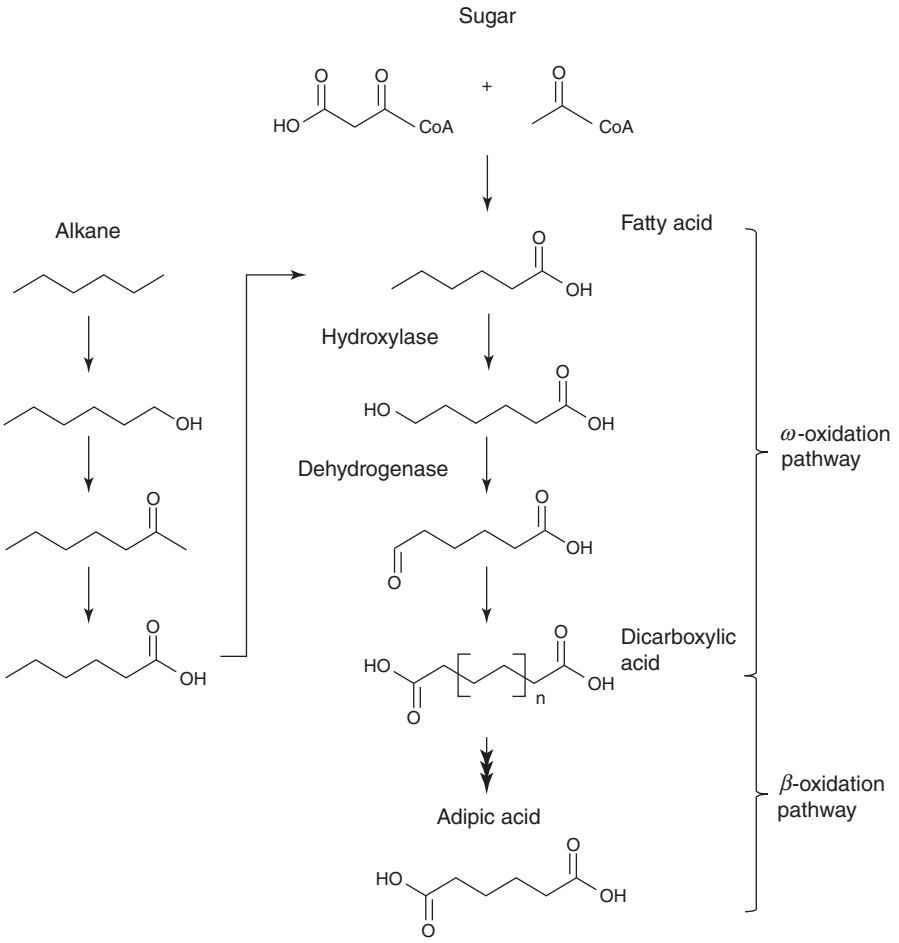



**FIGURE 19.2** Verdezyne process for production of bio-adipic acid from sugars, fats, and alkanes with *Candia tropicalis* via the ω-oxidation pathway (Picataggio et al., 1992; Picataggio and Beardslee, 2012). When alkanes or fatty acids are used as feedstock, these typically long-chain substrates are depending on the number of carbons, repeatedly shortened via the β-oxidation pathway to obtain bio-adipic acid.

technologies have been developed to depolymerize lignin to valuable products such as small aromatic compounds (Bu et al., 2011; Ma et al., 2012). So far, it is difficult to directly produce commercial chemicals from lignin at sufficient purity. For a validation protocol the complex structure of lignin needs to be understood to predict the mechanisms of depolymerization. An additional complicating factor is the variety on aromatic chemical composition of different plant species. These problems can be overcome by application of fermentation, since microorganisms can use crude/impure compounds as substrate.

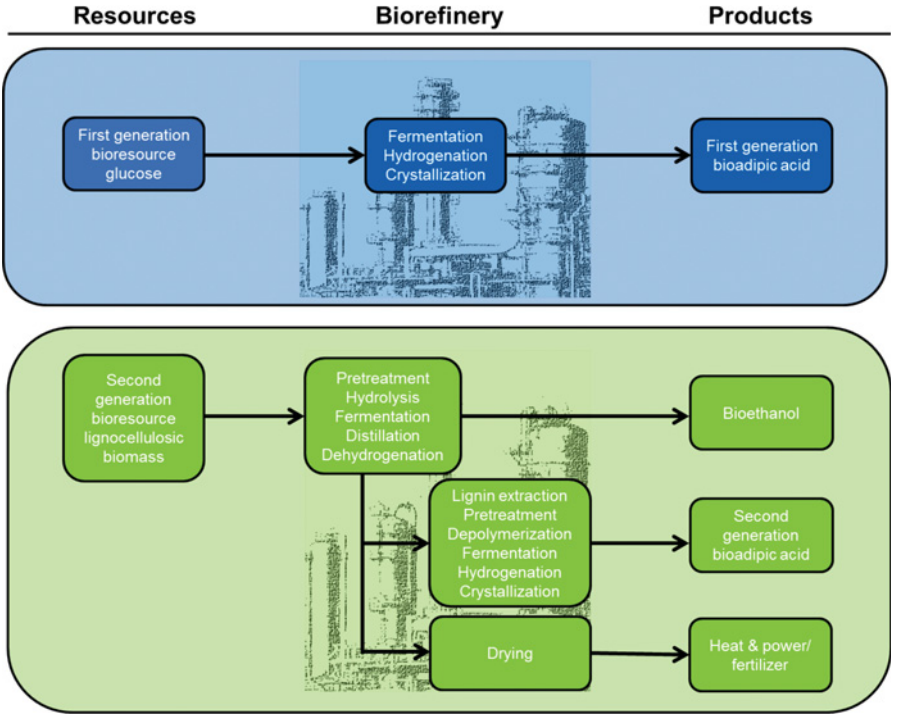



**FIGURE 19.3** Biorefinery concepts that integrate the first and second generation bio-adipic acid. (*For a color version, see the color plate section.*)

## 19.2 PRODUCTION OF BIO-ADIPIC ACID

### 19.2.1 Natural Formation by Microorganisms

Bio-adipic acid is formed by different bacteria (Cheng et al., 2002). As example, it results from naturally occurring degradation of cyclohexane, cyclohexanol, and cyclohexanone (Figure 19.4).

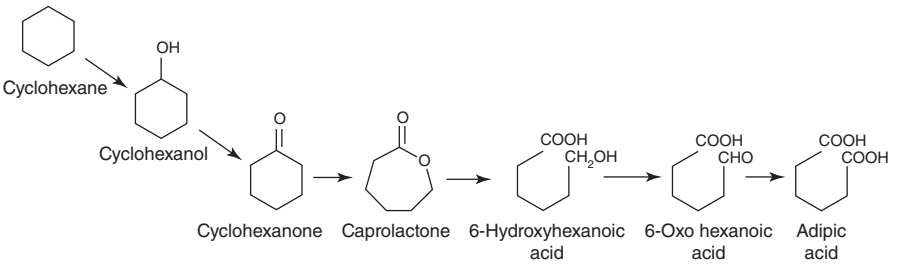



**FIGURE 19.4** Natural degradation of cyclohexane to adipic acid by microorganisms (Cheng et al., 2002).

Relevant metabolic genes and enzymes from different species have been analyzed and tested for activity in recent years (Polen et al., 2012). The corresponding pathways from *Acinetobacter* sp., *Arthrobacter* sp., or *Rodococcus* sp. have been found functional in *Escherichia coli* as heterologous host (Cheng et al., 2000; Brzostowicz et al., 2003). These mutants indeed accumulated bio-adipic acid from cyclohexanol, although at low amount and rate, far below a commercially attractive level. As recently reviewed, no natural metabolic pathway, discovered so far, allows efficient biotechnological production (Polen et al., 2012) so that new concepts from systems metabolic engineering and synthetic pathway engineering are needed for sustainable production of bio-adipic acid (Becker and Wittmann, 2012; Wittmann and Lee, 2012).

### 19.2.2 First Generation Bio-Adipic Acid

***19.2.2.1 Biotechnological De-Novo Biosynthesis From Glucose*** Glucose, the major traditional raw material in industrial biotechnology, can be used as renewable feedstock for bio-adipic acid. However, the structure of the sugar substrate is different from that of the product, so that the biotechnological conversion is complex, involves a cascade of enzymatic steps, and requires efficient cofactor regeneration. Particularly, *Candida tropicalis* has been engineered into an excellent whole-cell biocatalyst for direct production of bio-adipic acid from glucose (Picataggio et al., 1992; Picataggio and Beardslee, 2012) (Figure 19.2). The yeast is successfully applied in different biotechnological processes and able to grow at low pH. Such a characteristic is very important for the development of a continuous process. *cis*,*cis*-Muconic acid precipitates at a pH below its second pKa value of 3.03, which enables cell and medium recycling (Leong, 2007). The genome of *C. tropicalis* has been sequenced and largely annotated, which provides a rich source of knowledge for its targeted genetic modification (Butler et al., 2009). Important improvements of the yeast's performance have been achieved through metabolic engineering of its fatty acid metabolism. This involved stepwise reduction of the activity of the β-oxidation pathway by subsequent deletion of the four genes that encode for both isozymes of acetyl-CoA oxidase, catalyzing the first reaction of the β-oxidation pathway. As consequence, glucose is converted into bio-adipic acid via acetyl-CoA through fatty acid synthesis and the ω-oxidation pathway. This reaction sequence forms fatty acids as intermediates from glucose, which are then further processed into the final product. Hereby, a hydroxyl group is introduced at the ω-carbon of the fatty acid and then oxidized into a carboxyl group. The hydroxyl group is catalyzed by two successive oxidation steps, which involve alcohol dehydrogenase and aldehyde dehydrogenase. Among other modifications, debottlenecking of ω-hydroxylation by amplifying the genes encoding for cytochrome P450 monooxygenase and NADPH-cytochrome reductase was found crucial for efficient production. An attractive molar yield of about 40% from glucose has been claimed in certain embodiments (Picataggio and Beardslee, 2012). A production process has been developed on basis of these designer mutants by the company Verdezyne. Hereby, the use of fatty acids as pathway intermediates opens the possibility to widen the substrate spectrum of this process toward fats, vegetable oils, and alkanes, as non-food raw materials. Depending

on the number of carbons in these typically long-chain substrates, this requires the β-oxidation pathway, of a partially β-oxidation blocked *C. tropicalis* strain, to repeatedly shorten the carbon backbone and to accumulate bio-adipic acid. At present, this route competes with the chemical process from glucose, operated by Rennovia (Figure 19.1) (Boussie et al., 2010; de Guzman, 2010; Dapsens et al., 2012). The chemical alternative from glucose claims a molar yield of 66% for the first step to glucaric acid and of 89% for the second step to bio-adipic acid (Boussie et al., 2010).

***19.2.2.2 Combined Biotechnological–Chemical Synthesis From Glucose*** An interesting concept toward bio-adipic acids comprises a combined strategy that links initial biochemical conversion of glucose into the intermediate *cis*,*cis*-muconic acid with subsequent chemical reduction into bio-adipic acid. In the first step, *cis*,*cis*-muconic acid is formed under aerobic conditions at pH 6.3 and 36°C from glucose by heterologous *E. coli* (Draths and Frost, 1993; Niu et al., 2002). Given interesting process advantages, including lower pH values, a similar heterologous strain was developed with *Saccharomyces cerevisiae* (Curran et al., 2013; Weber et al., 2012). Acid resistance may also be possible for *E. coli*, *P. putida*, and *Bacillus subtilis* through heterologous gene expression and directed evolution (Guazzaroni et al., 2013). This novel route is based on a synthetic pathway that links aromatic amino acid biosynthesis with catalytic steps for aromatization of the intermediate 3-dehydroshikimic acid and its further conversion via protocatechuate and catechol into the final product (Figure 19.5).

The heterologous genes that were introduced into in *E. coli* and *S. cerevisiae* comprise 3-dehydroshikimic acid dehydratase, protocatechuic acid decarboxylase, and catechol 1,2-dioxygenase. In *E. coli*, the first two genes were expressed from *Klebsiella pneumoniae* strain A170-40 and the last gene from *Acinetobacter calcoaceticus*. For this strategy, a molar yield of 22% *cis*,*cis*-muconic acid from glucose was reached (Table 19.1). The intermediate accumulated to about 37 g/L and could be easily hydrogenated at a yield of 90% after addition of catalytic 10% platinum on carbon at a low elevated pressure of 3.5 bars within 3 hours. The three heterologous genes integrated into *S. cerevisiae* are genes from *Podospora anserine*, *Enterobacter cloacae*, and *Candida albicans*, respectively. Together with other modifications of the metabolism enabled the production of *cis*,*cis*-muconic acid with *S. cerevisiae* at a molar yield of 0.5% *cis*,*cis*-muconic acid and the intermediate accumulated to about 0.14 g/L. Another new artificial biosynthetic route has been described for *E. coli* that uses besides glucose also glycerol as substrate. Anthranilic acid, the first intermediate in the blocked tryptophan biosynthetic branch, was subsequently converted to catechol and *cis*,*cis*-muconic acid (Sun et al., 2013). Genes for anthranilate 1,2-dioxygenase and catechol 1,2-dioxygenase were coexpressed from *P. aeruginosa* and *P. putida* KT2440, respectively. A maximum titer of 0.39 g/L of *cis*,*cis*-muconic acid and a molar yield of 2.2% was described for flask experiments. In addition, an alternative strategy via glucaric acid as intermediate has been proposed, which is based on three heterologous expressed genes in *E. coli* encoding myo-inositol-1-phosphate synthase, myo-inositol oxygenase, and uronate dehydrogenase from *S. cerevisiae*, mice and *Pseudomonas syringae*, respectively (Moon et al., 2009). At present, the

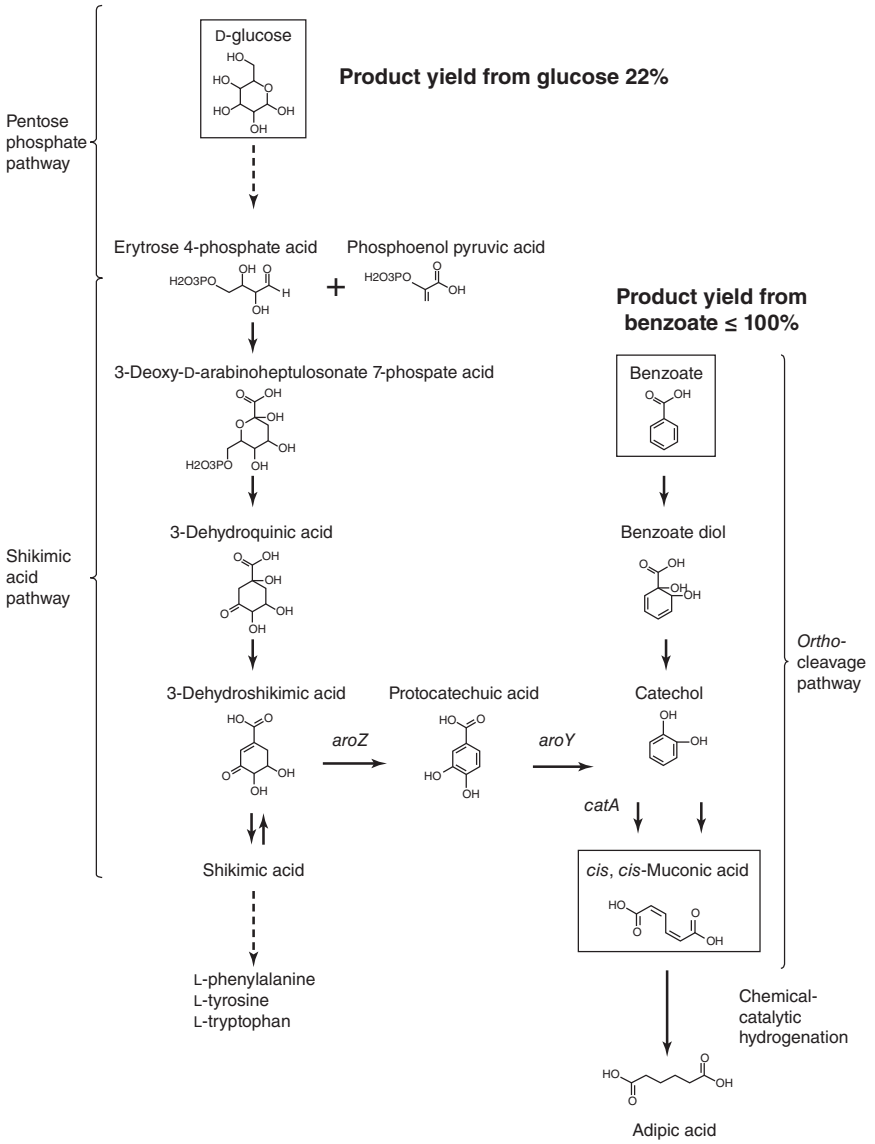



**FIGURE 19.5** Combined biotechnological–chemical synthesis of bio-adipic acid from glucose. Via heterologous expressed genes in *Escherichia coli* and *Saccharomyces cerevisiae* glucose was converted to *cis*,*cis*-muconic acid at a molar yield of 22% and 0.5%, respectively, via heterologous expression of genes [*aroZ* (3-dehydroshikimic acid dehydratase), *aroY* (protocatechuic acid decarboxylase), and *catA* (catechol 1,2-dioxygenase)] (Draths and Frost, 1993; Niu et al., 2002, Curran *et al.*, 2013). *Arthrobacter* sp. and *Pseudomonas* sp. are able to convert benzoate to *cis*,*cis*-muconic acid at a molar yield close to 100% (Schmidt and Knackmuss, 1984; Mizuno et al., 1988; Chua and Hsieh, 1990; Bang and Choi, 1995; Choi et al., 1997; van Duuren et al., 2012). Subsequently, *cis*,*cis*-muconic acid is converted to bio-adipic acid by chemical catalytic hydrogenation.

**TABLE 19.1 Biotechnological Production of *Cis*,*Cis*-Muconic Acid**

| Process | Strain | Substrate | Maximum specific productivity (g/g-dcw·h) | Maximum volumetric productivity (g/L·h) | Final concentration (g/L) | Molar yield % |
|---|---|---|---|---|---|---|
| pH-stat fed-batch | *Pseudomonas putida* KT2440-JD1 | Benzoate | 0.60 | 0.8 | 18.5 | ≤100 |
| Cell-recycle system[a] | *P. putida* BM014 | | 0.14 | 5.5 | 12.0 | 73 |
| DO-stat fed-batch | *P. putida* BM014 | | 0.21 | 2.2 | 32.0 | ≤100 |
| Fed-batch | *Arthrobacter* sp. | | nd | 1.1 | 44.0 | 96 |
| Fed-batch | *Pseudomonas* sp. B13 | | 0.24 | 0.8 | 7.0 | 91 |
| Fed-batch | *Pseudomonas* sp. | Toluene | nd | nd | 45.0 | nd |
| Shake flask cultures | *E. coli* BW25113 | ΔtrpD::kan Glucose and Glycerol | nd | $3 \times 10^{-3}$ | 0.39 | 2.2 |
| DO-stat fed-batch | *Escherichia coli* WN1/pWN2.248 | Glucose | 0.05 | 1.3 | 36.8 | 22 |
| Shake flask culture | *Saccharomyces cerevisiae* | | nd | $8.3 \times 10^{-3}$ | $141 \times 10^{-3}$ | 0.4 |

The data given comprise maximum specific productivity, maximum volumetric productivity, final concentration, and molar yield and are taken from previous work (Schmidt and Knackmuss, 1984; Mizuno et al., 1988; Chua and Hsieh, 1990; Bang and Choi, 1995; Choi et al., 1997; Niu et al., 2002; van Duuren et al., 2012; Curran et al., 2013; Sun at al., 2013)
nd, not determined.
[a]Steady values.

production of bio-adipic acid via glucaric acid seems not efficient enough for a commercial process. An alternative synthetic pathway for the production of this compound is proposed via glucodialdose (Lippow et al., 2010).

### 19.2.3 Second Generation Bio-Adipic Acid

Beyond sugar-based production, non-food raw materials appear particularly promising for bio-adipic acid. A first attempt into this direction is the extension of the substrate spectrum for recombinant *C. tropicalis* to fats and oils (see 19.2.2.1). As mentioned, the chemical structure of these substrates differs substantially from that of the target product, which requires a comprehensive pathway set for production. In this regard low-value aromatic compounds from lignin seem interesting alternative feedstocks. Naturally, the β-ketoadipate pathway serves for the convergent conversion of such aromatic compounds derived from lignin and other plant materials (Harwood and Parales, 1996). Additionally to this pathway, phenyl acetic acid and homogentisic acid are central aromatic metabolites (Jiménez et al., 2002). The ability to convert these compounds can be used for the degradation of some aromatic pollutants. The β-ketoadipate pathway is widely distributed among the microbial world and is present, for example, in *Pseudomonas putida*, *Acinetobacter calcoaceticus*, *Agrobacterium tumefaciens*, and *Rodococcus erythropolis* (Harwood and Parales, 1996). These exhibit differences in gene organization and the enzymatic set. Depending on the nature of starting aromatic molecule, the pathway converges at catechol or at protocatechuic acid as central metabolite (Figure 19.6). As example, quinic acid, vanillic acid, 4-coumaric acid, ferulic acid, and coniferyl alcohol are converted to protocatechuic acid. Unsubstituted aromatic compounds such as naphthalene, benzene, anthranilic acid, toluene, and phenol are metabolized via the catechol branch. In the latter cases, this gives direct access to *cis*,*cis*-muconic acid, an intermediate directly downstream in the β-ketoadipate pathway (Figure 19.6). The aromatic compounds that are not converted via catechol can be used as feedstock for growth and maintenance.

This available biocatalytic potential has been exploited since many years by selecting classical mutants with beneficial phenotypes. Different microorganisms accumulate *cis*,*cis*-muconic acid from benzoate or toluene, as model substrates for small aromatic compounds, examples being *Pseudomonas* sp. and *Arthrobacter* sp. (Schmidt and Knackmuss, 1984; Mizuno et al., 1988; Chua and Hsieh, 1990; Bang and Choi, 1995; Choi et al., 1997; van Duuren et al., 2012) (Table 19.1). As explained above, the intermediate can be easily hydrogenated to yield bio-adipic acid. Overall, surprisingly high titers for *cis*,*cis*-muconic acid of more than 40 g/L have been reported. Hereby, *cis*,*cis*-muconic acid was accumulated at a molar yield close to 100%. Such a high yield is enabled by the fact that the conversion is independent from the central metabolism. For pure benzoate and toluene, the process was successfully scaled to 500 and 1500 L, respectively (Table 19.1). Sophisticated bioprocess development further improved the process performance. A precise process control is rather important to avoid the accumulation of catechol, because it is toxic for the cells (Schreiber et al., 1980; Schmidt and Knackmuss, 1984; Benndorf et al., 2001; Benndorf et al., 2006).

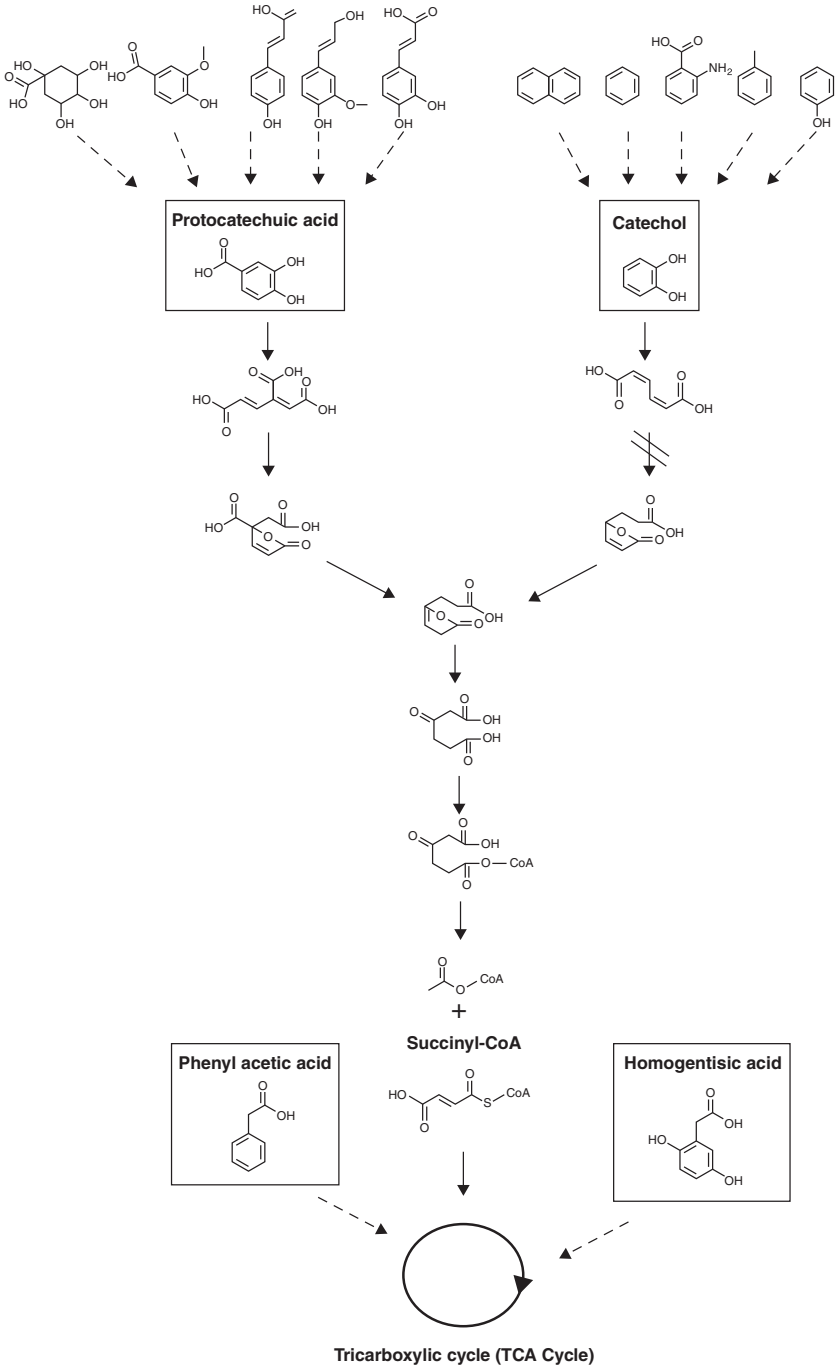



**FIGURE 19.6** The two branches of the β-ketoadipate pathway, which comprise catechol and protocatechuic acid as central metabolites (Harwood and Parales, 1996). To obtain *cis*,*cis*-muconic acid its further conversion has to be blocked. Additionally, other small aromatic compounds are converted via different pathways of which phenyl acetic acid and homogentisic acid are central metabolites (Jiménez et al., 2002).

This is complicated by the fact that the conversion of catechol to *cis*,*cis*-muconic acid is considered rate-limiting (Kim et al., 1988).

For efficient accumulation of *cis*,*cis*-muconic acid, a pH-stat fed-batch and a dissolved oxygen (DO)-stat fed-batch approach were developed to avoid the accumulation of toxic intermediate metabolites (Bang and Choi, 1995; Niu et al., 2002; van Duuren et al., 2012). The pH strategy directly couples the microbial activity to the feed rate. When benzoate is converted to *cis*,*cis*-muconic acid, the medium acidifies in stoichiometric amounts. This enables to couple the feeding of benzoate directly to the formation of *cis*,*cis*-muconic acid via the pH control. Accordingly, the feed rate is automatically set optimal for production, even when the production is unexpectedly reduced. The pH-stat fed-batch process was found reliable, given the uncontrolled catechol accumulation, when glucose was fed on basis of the dissolved oxygen level. During the pH-stat fed-batch process, a specific production rate of 0.6 g/g·h was obtained, which is higher than values of other process (Table 19.1). For the future, rational metabolic engineering strategies, available in the meantime, promise a next level of producing strains. It might be possible to reach a final concentration of up to 80 g/L of *cis*,*cis*-muconic acid (van Duuren et al., 2012).

## 19.3 ECOLOGICAL FOOTPRINT OF BIO-ADIPIC ACID

To evaluate the ecological footprint of bio-adipic acid and to compare it to that of conventional petrochemistry, an extended life cycle assessment (LCA) was recently performed (van Duuren et al., 2011). Given that the kind of feedstock for production is determining the most costly portion of the ecological footprint, the assessment was taken further by additionally incorporating the glucose-based process for comparison (glucose 1 Figure 19.7). This focused on the investigation of the combined biotechnological–chemical process from glucose to *cis*,*cis*-muconic acid by recombinant *E. coli*, where sufficient data were available to support a valid analysis (Niu et al., 2002) (Figure 19.7). To increase the efficiency of the process five recycle times were considered. To evaluate the relevance of cell and medium recycling for the LCA, by conducting a fermentation at low pH, the energy demand and $CO_2$eq emission for medium preparation were divided five times (glucose 2 in Figure 19.7). To cover the increased biomass concentration the quantity of added glucose and ammonium sulphate remained unchanged per ton product as under fed-batch conditions. Low pH causes stress for microorganisms, which negative effect was assumed to be compensated during the process by high cell density. Furthermore, the impact on the downstream process was expected to be negligible. To recycle cells after micro-filtration an extra cross-flow filtration step is applied. Besides the extra effort, this technique is also expected to improve the conditions for hydrogenation and evaporation due to a higher final concentration and the use of solvents with a lower boiling point.

As basis for the assessment, the ISO14040 series for environmental management systems were followed to a great extent for the CED and the $CO_2$ equivalent ($CO_2$eq) emission ($CO_2$ and $N_2O$ measured separately) (ISO 14040:2006). The assessment

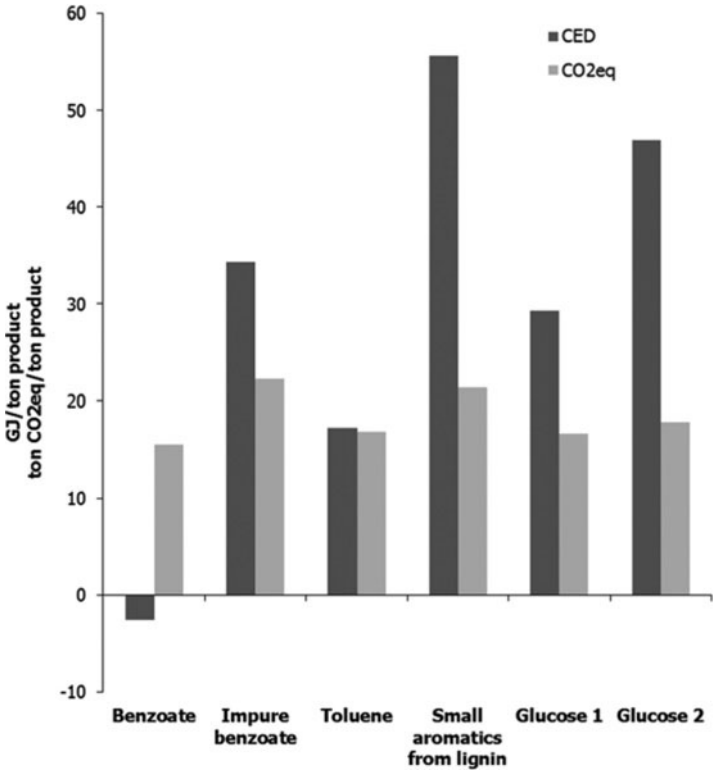



**FIGURE 19.7** Reduction of the cumulative energy demand (CED) and $CO_2$ equivalent ($CO_2$eq) emission per ton bio-adipic acid for the integrated biotechnological process using benzoate (62.4 GJ/ton, 6.8 ton/ton), impure benzoate (25.4 GJ/ton, no $CO_2$eq emission), toluene (56.4 GJ/ton, 7.2 ton/ton), small aromatics from lignin (5.1 GJ/ton, 1.2 ton/ton) and glucose (5.1 GJ/ton, 1.0 ton/ton) as feedstock in relation to the petrochemical process (104.24 GJ/ton, 25.9 ton/ton). To compare the LCA outcome for aromatic compounds and glucose as feedstock, a final concentration of 37 g/L was assumed, given this concentration was reached by *Escherichia coli* with glucose as feedstock (Niu et al., 2002). The medium and the process were kept similar. Glucose 1 represents a process without cell and medium recycling, Glucose 2 represents a process in which the biomass and medium are recycled 5 times.

was started at the point of fossil fuel delivery as naphtha products and natural gas. Furthermore, the assessment was defined till the factory gate for nylon-6,6 production, which is essentially the delivery point of pure bio-adipic acid. For $CO_2$ emission relating to the formation of benzoate and toluene, the assessment was taken beyond the factory gate to the point of incineration. Because impure benzoate represents a waste product from the BTEX naphtha cracker, solely its intrinsic energy value was taken into account. Phenol, as model substrate, represents impure small aromatic compounds from lignin. For its formation derived from wheat, the theoretical energy demand was based on the associated agricultural system and the dry weight of biomass

(Brehmer, 2008; Brehmer et al., 2009). Via pyrolysis a molar yield of 40% small aromatics can be generated from lignocellulosic biomass and directly form lignin, which are converted biotechnologically to *cis*,*cis*-muconic acid (Bu et al., 2011; Ma et al., 2012). The energy required for this process step is provided by the combustion of pyrolysis gases. Lignin is considered as a waste product from the production of the second generation raw materials cellulose and hemicellulose, because it is not the primary product for which plants are grown and processed (Tuck et al., 2012). The physicochemical formation of lignin from lignocellulose was therefore not eligible for the LCA given these process steps are calculated for the primary product formation. The production of glucose was based on the saccharide content of whole sugar beets. A similar approach for glucose was selected, as was done for lignin to calculate the theoretical energy demand and greenhouse gas emission. However, additionally 2 GJ/ton was encountered for the process. Furthermore, a molar yield of 96% and 22% *cis*,*cis*-muconic acid was considered, when small aromatic compounds or glucose were used as feedstock, respectively (Bang and Choi, 1995; Choi et al., 1997; Schmidt and Knackmuss, 1984; Mizuno et al., 1988; Chua and Hsieh, 1990; Niu et al., 2002; van Duuren et al., 2012). For the chemical catalytic hydrogenation step of *cis*,*cis*-muconic acid to bio-adipic acid, a slightly increased pressure (3.5 bar) and a yield of 90% was taken into account. The $CO_2$ emission was simulated by multiplying the energy use with the emission factor of heating oil as defined by the National Greenhouse Accounts Factors of the Department of Climate Change and Energy Efficiency of the Australian Government. The $N_2O$ emission related to the use of fertilizers was included in the LCA. Namely, by agricultural soil management and nitric acid formation for the production of the fertilizer NPK-CAN, $N_2O$ is produced (Nevison, 2002; Brehmer et al., 2008; Neuffer et al., 2010). Possibly in the future, the negative effect of nitric acid production can be reduced by the application of *Rhizobium* symbiosis to non-*papilionaceous* plants for the fixation of nitrogen (Ivanov et al., 2012).

Without medium reuse, 27 L of water are required for 1 kg product at a titer of 37 g/L *cis*,*cis*-muconic acid. The environmental impact of two times autoclaving 1 ton of water equals 1.4 GJ and 95 kg $CO_2$eq, respectively. Only one the two autoclavation steps was incorporated in the LCA to compensate for the intrinsic energy value of biomass that can be further used in closed loop processes. For this calculation, clear water treatment and waste water treatment was also taken into account, but it encompasses only less than one percent of the calculated energy demand (Vairavamoorthy, 2011). Furthermore, pH regulation with HCl (4.2 GJ/ton, 0.2 ton/ton) was included at a feed level of 0.58 ton/ton. To offset the ammonium use, it is interesting to feed the protein-rich coproduct slurry of the glucose production during the fermentation step.

Bio-adipic acid from lignin, revealed a 53% reduction in the total energy demand as compared to classical adipic acid from the petrochemical route (Figure 19.7). Moreover, the greenhouse gas emission was reduced even by 83%. In addition, the glucose-based process exhibited a better ecological footprint than that from traditional petrochemistry, although it did not reach the excellent performance of the lignin-based process. However, when cell and medium recycling is applied with glucose as substrate the environmental impact is competitive.

Glucose as feedstock for production of *cis*,*cis*-muconic acid was added to model the present leading industrial business concepts. However, since the processes applied by Verdezyne and Rennovia have a different setup, they cannot directly be represented by this LCA. Furthermore, the ecological and commercial competitiveness of these two processes is mainly dependent on undefined parameters of the processes, for example, product yield, product rate, and final concentration. In view of the good properties of glucose as feedstock, the efficiency of the process is expected to be competitive compared to the conventional petrochemical process. To give an impression of how the yield and product concentration of *cis*,*cis*-muconic acid influence the environmental impact, a molar yield of 25–100% and a concentration of 25–100 g/L were taken into account in Figure 19.8. When glucose is used for production, as well as for growth and maintenance, the theoretical molar yield of 100% is unattainable. At the same yield and titer for glucose or impure small aromatic compounds as feedstock, the environmental impact has a fairly uniform development. The process of Verdezyne does not apply any chemical catalytic step, because it converts glucose, fats, and alkanes directly to bio-adipic acid (Picataggio et al., 1992; Picataggio and Beardslee, 2012) (Figure 19.2). This possibly results in significantly less energy demand for the process given the chemical catalytic hydrogenation step is the most energy demanding step. The process of Rennovia applies an extra chemical catalytic step concerning the oxidation of glucose to glucaric acid (Boussie et al., 2010; de Guzman, 2010; Dapsens et al., 2012) (Figure 19.1). Although chemical catalytic steps cost a lot of energy, they also might be more efficient compared to a fermentative conversion given the conditions can be optimally chosen. By cell and medium recycling the energy demand for the process was reduced up to 60% and 50% for the substrates glucose and lignin, respectively, and the $CO_2$eq emission was reduced by a maximum of 10% for both substrates at 25 g/L and 25% molar yield. At 100 g/L the decrease of energy demand was only 10% and $CO_2$eq emission remained the same for both substrates. The reduction in environmental impact increased exponentially with a decreasing titer. With a concentration of 25 g/L the decrease in energy demand was additionally negatively affected by an increasing molar yield in an exponential manner, leading to a reduction at 100% of only 40% GJ/ton for both substrates.

The great advantage of applying plant biomass as feedstock is that it is formed by photosynthesis and therefore low energy values for growth have to be taken into account. Even with a molar yield of 22% *cis*,*cis*-muconic acid from glucose, it is still possible to save energy as compared to the petrochemical production process of adipic acid (Figure 19.7). In comparison to second generation bioprocesses, first generation processes are in the majority, because organisms can easily use these carbon sources for growth and maintenance. By growing the species under limited conditions and/or by applying metabolic engineering, it becomes possible to accumulate metabolic intermediates including *cis*,*cis*-muconic acid (Polen et al., 2012). When the second generation feedstock lignin is used, the CED and $CO_2$eq emission are rather similar as for glucose (Figure 19.8). However, in particular because the conversion of small aromatics can lead to a high molar yield, the application of this second generation feedstock is very interesting. Therefore, based on this assessment, developed depolymerization techniques for lignin should be applied in industry as soon as possible to open the gate for the most sustainable bio-adipic acid.

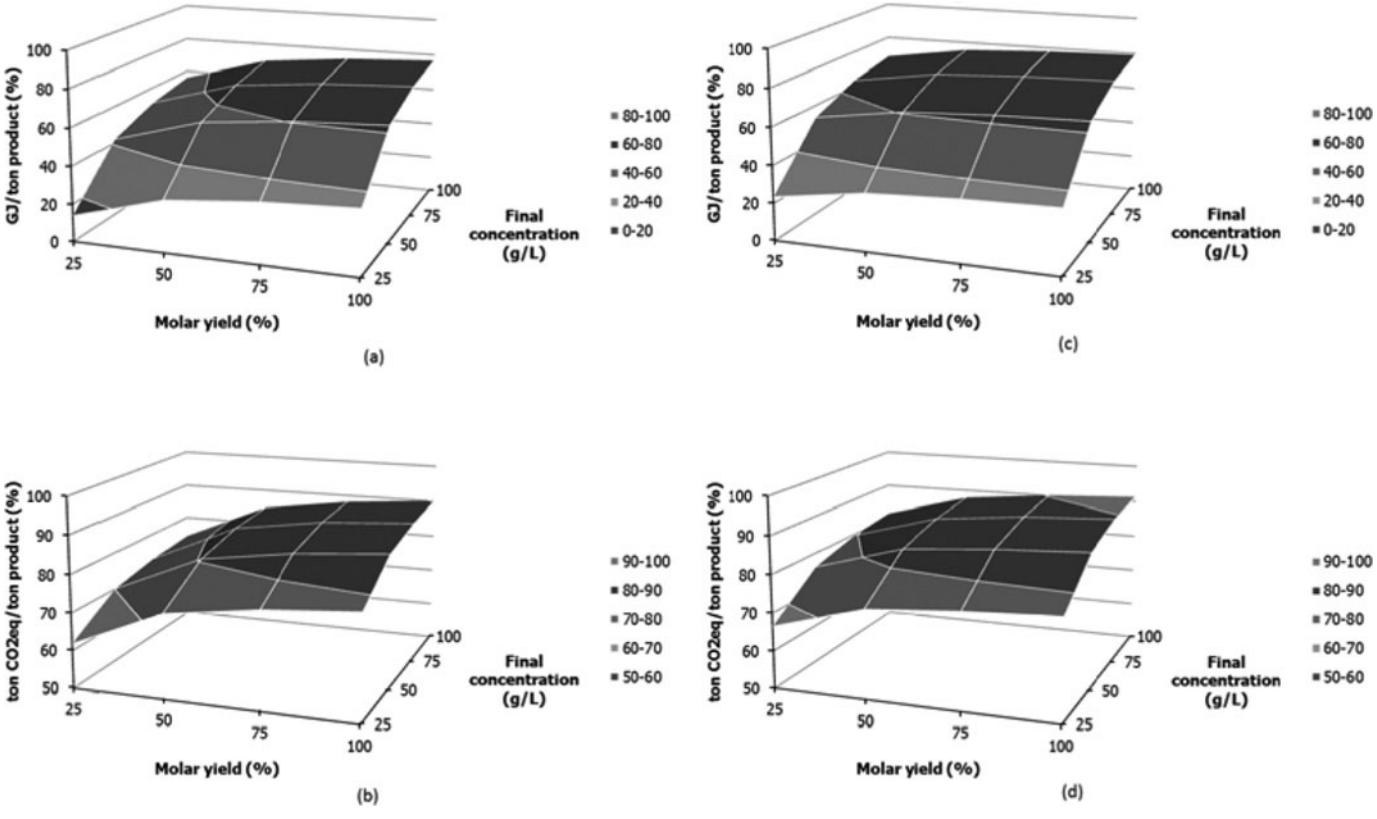



**FIGURE 19.8** Percentage of cumulative energy demand (a, c) and $CO_2$ equivalent emission (b, d) savings per ton bio-adipic acid at a molar yield of 25–100% and a concentration of 25–100 g/L of *cis*,*cis*-muconic acid for the integrated biotechnological process using glucose (a, b) and small aromatics from lignin (c, d) as feedstock compared to the petrochemical process.

## 19.4 ECONOMIC OUTLOOK

For an economic comparison of the various feedstocks to be used in an integrated biotechnological process, the costs per ton bio-adipic acid can be calculated. Table 19.2 lists the variable costs for a process with a final concentration of 37 g/L *cis*,*cis*-muconic acid in the fermentation broth. The variable costs encounter feedstocks, process, energy, and greenhouse gas emission costs. To estimate the price of lignin from wheat, €4/GJ is applied and a yield of 40% is taken into account (Reith et al., 2009). The intrinsic energy level of various second generation biomass resources does differ between 9.5 and 17.5 GJ/ton as defined by the European Biomass Industry Association. For wheat in particular, 14.5 GJ/ton was considered. When glucose is used as feedstock, the volatile price and relative low molar yield are points that require attention. Additionally, this substrate competes with the food supply, which should be avoided given the price-raising effect. Process costs in Table 19.2 include medium (trace elements, phosphate, ammonium, HCl) (197 Euro/ton), glucose as growth and maintenance substrate and the use of water (106 Euro/ton) based on Alibaba products, Indexmundi, and the price of water supply and sewage (BMU, 2011), respectively. When five recycle times were incorporated with glucose as substrate the cost were for medium formation 29 Euro/ton and for water use 21 Euro/ton. The medium formation was particularly reduced in price since pH regulation with HCl is not necessary at low pH. When glucose is converted to *cis*,*cis*-muconic acid, the process cost is lower compared to when small aromatic compounds are used, since no additional carbon source for growth and maintenance must be added. The energy costs and greenhouse gas emission are related to each step of the process. Given feedstocks are purchased, it is assumed that all costs are covered concerning the production, including the $CO_2$ emission costs. For 1 GJ of energy, €19.57 was calculated based on the price for heating oil (37.3 MJ/L) indicated by Europe's Energy Portal. Furthermore, for 1 ton $CO_2$eq emission €7.00 was calculated, which price

**TABLE 19.2 Variable Cost Price of the Production of Bio-Adipic Acid from Different Feedstocks Via an Integrated Biotechnological Process With *Cis*,*Cis*-Muconic Acid as Intermediate Metabolite at 37 g/L in the Fermentation Broth**

| | | | Variable cost price bio-adipic acid Euro/ton | | | | |
|---|---|---|---|---|---|---|---|
| Substrate | Euro/ton | Molar yield % | Feedstock | Process | Energy | Ghg | Total |
| Benzoate | 1188 | 0.86 | 1149 | 466 | 761 | 19 | 2394 |
| Impure benzoate | nd | | | | | | |
| Toluene | 951 | 0.86 | 694 | 466 | 761 | 19 | 1939 |
| Small aromatics | 145 | 0.86 | 108 | 466 | 761 | 19 | 1353 |
| Glucose 1 | 322 | 0.20 | 2005 | 303 | 761 | 19 | 3087 |
| Glucose 2 | 322 | 0.20 | 2005 | 50 | 468 | 12 | 2535 |

Ghg, greenhouse gas; nd, not determined. Glucose 1 represents a process without cell and medium recycling, Glucose 2 represents a process in which the biomass and medium are recycled 5 times.

was determined at Boerse.de. These two costs cover the energy use and greenhouse gas emission during the process, respectively. The process from small aromatic compounds is competitive in price compared to glucose given the low economic value of lignin and the high molar yield of *cis*,*cis*-muconic acid. It is important to note that the variable costs are not equal to the selling price. Besides profit, the fixed costs associated with labor, investment, housing, cleaning, and transportation as well as other costs, for example, for licensing, sale, logistics, marketing, and R&D should also be encountered. These costs should be handled with care, as dependent on individual cases. Figure 19.9 displays the impact of changing yield and titer. A low molar yield causes an increased demand for feedstock, an indirect extra demand for energy, and an extra emission of greenhouse gas. These effects increase the costs, that is, more money is spent on feedstock, energy, and $CO_2$-certificates to cover greenhouse gas emission. In order to allow economically attractive production (variable costs below the selling price of 1500 Euro/ton), at least for glucose, a yield of +/− 60% has to be achieved. Taking into account the effect of five recycle times the variable costs with glucose were reduced so that a yield of 40% is required. With cell and biomass recycling for glucose as well as lignin the variable costs decreased from 258 to 773 Euro/ton. The reduction in price increased exponentially with a decreasing titer. Based on the calculation, the variable cost price with lignin is under the most relevant circumstances under €1500. Given the favorable conditions for lignin, the use of this substrate can lead to a very successful second generation process for the production of bio-adipic acid.

## 19.5 FUTURE PROSPECTS

Petrochemical production of adipic acid has a high, unfavorable ecological footprint (Vink et al., 2003). As a result, there is a great potential for sustainable bio-based alternatives. In recent years, this had led to substantial developments toward bio-adipic acid, which already have reached pilot scale operations by the companies Verdezyne and Rennovia. DSM, a large Dutch biotech company, has also indicated to add the compound to its portfolio. The future use of lignin-based raw materials promises an even higher potential. Raw materials for second-generation biofuels consist of 25% lignin, which displays a huge waste stream (de Wild et al., 2010; Tuck et al., 2012). To indicate the potential significance of this waste stream, 70% of oil is used for transport fuels. First second-generation lignocellulosic biorefineries at commercial scale (e.g. by ZeaChem, Süd-Chemie), recently been put into operation, cannot make efficient use of their lignin wastes. A possible use of this raw material for bio-adipic acid is therefore expected highly attractive. Additionally by providing an added value to this significant waste stream, the use of biomass becomes more profitable as a whole. This route might also supply products, other than bio-adipic acid. As first example, terephthalic acid is produced from *cis*,*cis*-muconic acid by Amyris in the United States. An extended use of *cis*,*cis*-muconic acid is currently limited by the rather high price, but markets could grow, when the price would be lowered. In particular, the second generation integrated-biotechnological process

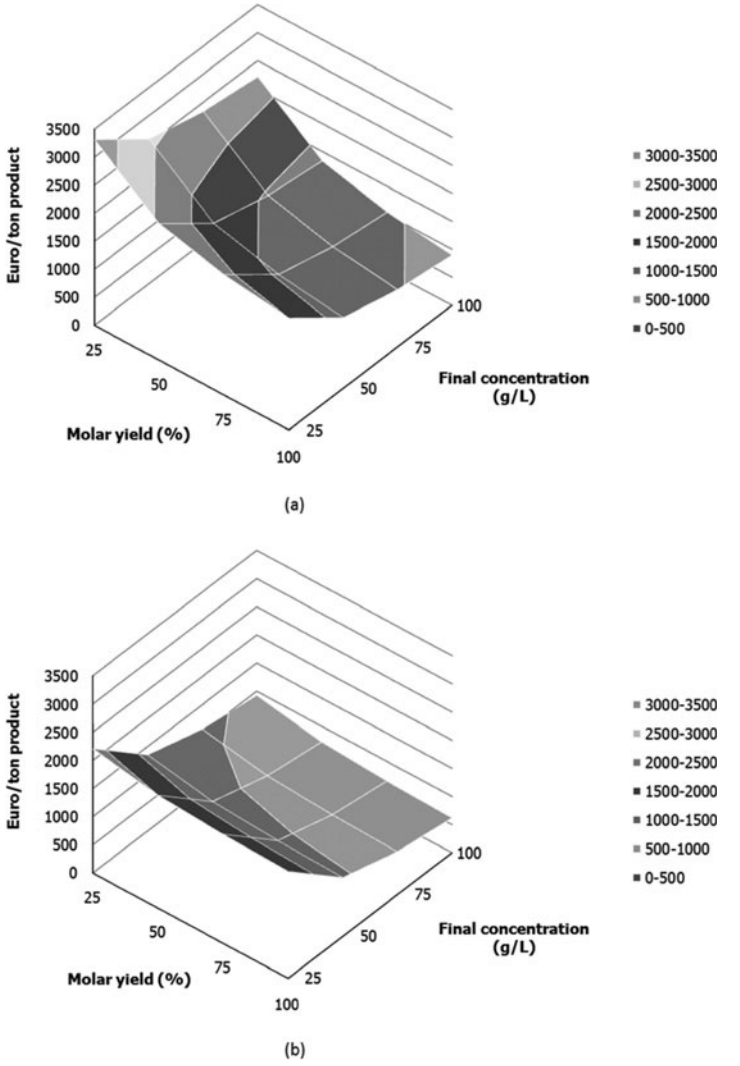



**FIGURE 19.9** The variable cost price in euros per ton bio-adipic acid at a molar yield of 25–100% and a concentration of 25–100 g/L of *cis*,*cis*-muconic acid when glucose (A) or impure small aromatic compounds from lignin (B) are used as feedstock.

for bio-adipic acid, which uses lignin as substrate, can be seen as a trendsetter for the production of bio-based bulk chemicals. Important tasks remaining, involve the validation on pilot scale how lignin can best be depolymerized for industrial integrated-biotechnological processes. Moreover, to ensure a low environmental impact under such circumstances, it is important to establish a high molar yield and final concentration with the integrated biotechnological process.

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